

Project Description – Project Proposals in the Area of Scientific Library Services and Information Systems (LIS)

LIS Funding Programme or Call:

e-Research-Technologies

[Project title]

NORA – Building a Sustainable, Interoperable Infrastructure for Medical Imaging Research

Oder NORA - Next-generation Open Research Platform for Medical Image Analysis

[First name last name, city]

Project Description

Sections 1-3 must not exceed 15 pages in total.

1 Starting Point

State of the art and preliminary work

Medical imaging is a cornerstone of modern medicine. It enables physicians to diagnose disease, plan therapy, and monitor outcomes, and it represents one of the most important sources of data in biomedical research. Advances in imaging technologies and computational analysis have transformed the field into a highly interdisciplinary domain that combines medicine, physics, computer science, statistics, and increasingly artificial intelligence (AI) to extract knowledge from complex image data.

This development creates major opportunities but also substantial challenges. Modern medical imaging research operates on a scale and complexity that far exceed traditional tool-based analyses. Large and heterogeneous datasets originating from different scanners, protocols, institutions, and patient populations must be curated, processed, and analysed in a standardized and reproducible manner. This challenge is further amplified by population-based studies, multi-centre collaborations, and the increasing use of AI methods, which require tightly integrated workflows spanning data access, annotation, algorithm integration, large-scale processing, evaluation, and the linkage of imaging data with structured and unstructured clinical information.

Despite this clear need, no comprehensive infrastructure has yet become established for managing these complex workflows in medical imaging research. Existing free and commercial tools typically focus on individual steps of the workflow—such as image viewing, annotation, or standalone algorithm development—and must therefore be combined into fragile, project-specific pipelines. Researchers often rely on fragmented tools and ad-hoc solutions that cannot scale, are difficult to reproduce, and make collaboration cumbersome.

As a result, AI developments in medical imaging frequently remain isolated within individual projects and only rarely translate into clinical routine.

The current lack of integrated systems therefore results in substantial inefficiencies, hampers reproducibility, and complicates collaborative work — precisely at a time when medical imaging research increasingly depends on scalable infrastructures capable of exploiting large imaging datasets and modern AI methods. Overcoming this fragmentation is thus one of the most urgent prerequisites for translating advances in imaging analysis into tangible medical progress.

Preliminary Work

The applicants form an interdisciplinary team of software engineers, computer scientists, imaging scientists, and clinicians at the University Medical Center Freiburg with long-standing experience in medical image analysis across methodological development, clinical applications, and large-scale data-driven studies. Over many years, the team has developed algorithms, software tools, and computational workflows for a wide range of imaging modalities and research questions, and has coordinated numerous interdisciplinary projects that integrate medical expertise with advanced computational methods.

Confronted with the needs of everyday clinical and research practice, the team initiated the development of NORA (*Next-generation Open Research Platform for Medical Image Analysis*). Unlike conceptual initiatives, NORA emerged directly from real clinical challenges and research workflows. Through continuous development, it has evolved into a powerful, practice-oriented platform that supports all relevant aspects of imaging research—from data management and annotation to large-scale data processing, AI integration, and evaluation. The platform demonstrates that medical imaging research at unprecedented scale is possible within a single, integrated system: with only a few interactions, users can query PACS data, train and apply AI models on tens of thousands cases, perform large-scale cluster processing, and derive quality-controlled, metadata-linked results in a multi-user environment.

NORA is used intensively within our institution, serving as the backbone for numerous research projects and publications (currently more than 180 indexed in Google Scholar, **does this sound much?**) and has also become a hub for collaborations, connecting clinical departments, computer science groups, and external partners. The platform has attracted strong interest from the wider research community, and pilot installations at partner hospitals have demonstrated both feasibility and demand. At present, however, the groups' development efforts are primarily devoted to maintaining and extending core functionality and to supporting internal and selected partner installations. While this focus has ensured a high level of robustness and adaptability, it also means that broader and external usability have not yet been systematically addressed. Furthermore, the stepwise development in response to diverse research demands has given NORA remarkable flexibility and functionality, but at the same time, this organic growth has led to a certain complexity, which currently limits broad external deployment.

The goal of this proposal is to consolidate NORA and prepare it for broad community use. By strengthening stability, modularity, and accessibility, and by enabling the integration of external toolboxes and collaborative developments, NORA will evolve from a local pilot into a

sustainable research infrastructure for reproducible, scalable, and collaborative medical imaging research, positioned to become a nationwide and internationally adopted platform for open medical imaging research.

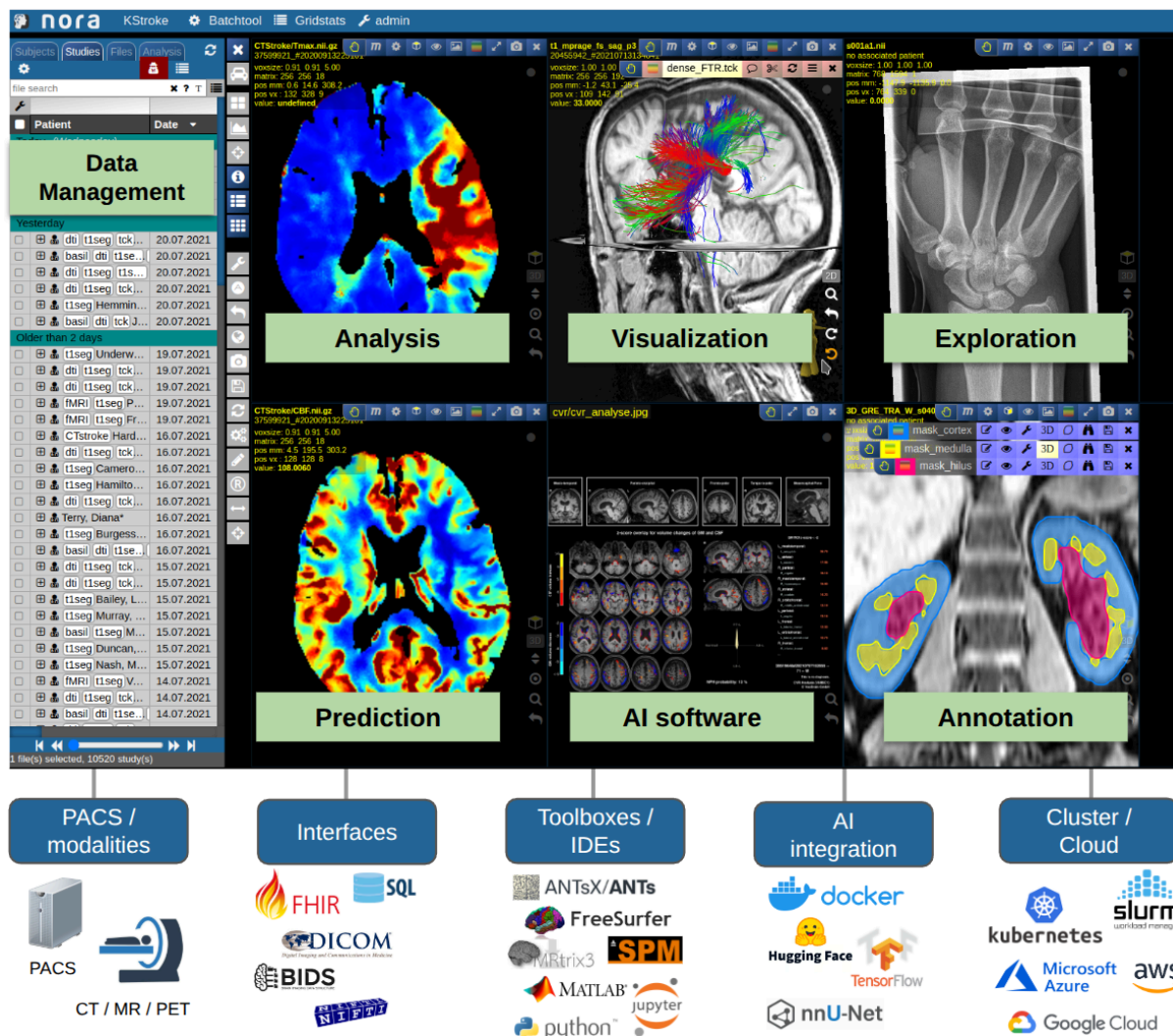


Figure 1: Extra Figure für backend / architecture?

NORA provides a fully web-based environment for medical imaging research, enabling users to access, manage, and analyse data directly through the browser. The platform supports comprehensive data management functions including organisation, curation, and anonymisation of imaging datasets, with seamless connectors to PACS and other institutional archives and full support for common medical imaging formats and standards. Integrated viewing, interaction, and annotation tools allow manual, semi-automatic, and AI-assisted labelling, as well as exploration and quality control of images and derived results. This system supports robust evaluation workflows, including reproducibility checks, cross-validation, and benchmarking of algorithms. The platform integrates a wide range of external image processing toolboxes. For large studies and high-throughput research, NORA offers scalable batch processing with integrated job scheduling and resource allocation, enabling efficient execution of complex workflows across local or cloud computing.

2 Objectives and work programme

2.1 Anticipated total duration of the project

Please state the project's intended duration and how long DFG funds will be necessary, for ongoing projects: since when the project has been active.

The proposed project is planned for a total duration of three years. This period allows sufficient time for the stepwise implementation of all work packages described below.

2.2 Objectives

Please give a concise description of your project objective, relating it directly to the objectives given in the programme guidelines. Explain what specific results can be anticipated by optimising the information infrastructure, how they fit into existing national or international structures, and how these will benefit research. State to what extent the project should be seen as a model or a pilot project and what needs would be met beyond those of your own institution. Please indicate if you anticipate results that, as well as improving the information infrastructure, may be relevant from both research and non-research related perspectives (such as science policy, technology, the economy or society).

The objective of this project is to consolidate and further develop NORA (Next-generation Open Research Platform for Medical Image Analysis) into a robust, comprehensive, and broadly deployable research infrastructure for data-intensive medical imaging research. NORA integrates data management, annotation, algorithm development, high-performance computing, and large-scale AI analysis within a unified environment, enabling researchers to conduct reproducible imaging studies on thousands of datasets within collaborative workflows. The platform is designed to support modern AI methods, including large-scale machine learning and emerging approaches such as large language models for multimodal medical data analysis.

The project directly addresses the objectives of the DFG programme to strengthen research information infrastructures for data-intensive science. Rather than duplicating existing infrastructures, NORA complements national and international initiatives by providing a specialised environment for large-scale medical image analysis. Through standards-based interfaces and interoperable workflows, the platform is designed to integrate with existing data infrastructures and ecosystems, including initiatives within the National Research Data Infrastructure (NFDI), Netzwerk-Universitätsmedizin (NUM), and related European efforts.

Hier noch konkreter NFDI4health, NFDI4bioimage, RACoon?

The anticipated results include a modularised and stabilised software architecture, simplified deployment on institutional and high-performance computing systems, improved interoperability with external analysis toolboxes and clinical information systems, and enhanced usability for interdisciplinary research teams. These developments will enable large-scale imaging studies and AI-driven analyses to be conducted more efficiently and reproducibly, while lowering technical barriers for collaborative research.

The project is conceived as a model infrastructure for data-intensive imaging research that can be adopted beyond the University Medical Center Freiburg. By providing reusable software components, interoperable interfaces, and scalable analysis workflows, the resulting platform will address needs widely encountered in medical imaging research across institutions.

At the same time, the design explicitly accounts for constraints arising from data governance and heterogeneous hosting requirements in medical research. The platform therefore supports flexible deployment models ranging from locally controlled installations to distributed and cloud-based computing environments, ensuring that the resulting infrastructure remains usable under evolving regulatory and technical conditions.

Overall, the project will provide a comprehensive yet modular platform covering key components of medical image processing within a highly interoperable environment that complements existing tools and infrastructures. This will enable efficient large-scale studies while lowering technical barriers, allowing clinicians, students, and interdisciplinary researchers to work productively with complex imaging data. By supporting systematic validation of imaging algorithms and biomarkers, the platform will also facilitate the translation of methodological advances into clinical applications. The project directly aligns with the objectives of the DFG LIS e-Research programme by strengthening sustainable research information infrastructures for data-driven biomedical research.

2.3 Work programme and proposed methods

Please give a detailed account of the steps you are planning to take during the proposed funding period and describe the methods you will use. Indicate which methods are already available and what relevant standards and best practices must be observed. Also detail any assistance that will be required from outside of your working group or institution.

If relevant, explain any alternative approaches to the research methods. Where possible, explain any factors which, in your view, represent a risk to the implementation of the project. Where relevant, explain how you plan to avoid these risks and handle such situations. List the steps planned for each applicant and draw up a clear overview of the structure and a schedule of work for those participating in the project.

The quality of the work programme is critical to the success of a funding proposal. The work programme should clearly state how much funding will be requested, for what period, why the funds are needed and how they will be used, providing details on individual items or your own financial contributions under item 5.9 where applicable.

To achieve the objectives described above, the project is organised into four work packages covering platform consolidation, interoperability, reproducible deployment, and community adoption. The work programme builds on the existing NORA platform and prepares it for robust, modular, and broadly deployable use in medical imaging research.

Overview of work packages

- **WP 1 – Software Consolidation and Architecture:** WP1 consolidates the existing NORA code base and establishes a robust, modular, and maintainable software architecture. The work package provides the technical foundation for sustainable development and future extensions of the platform.
- **WP 2 – Interoperability and Integration** WP2 establishes interoperability with clinical systems, research infrastructures, and external analysis tools. The work package enables seamless integration of imaging workflows, algorithms, and clinical information within the NORA environment.
- **WP 3 – Reproducible Deployment and Infrastructure Integration:** WP3 enables reproducible deployment and operation of NORA across heterogeneous computing environments. The platform will support installation on institutional servers, research clusters, and cloud infrastructures, and will be validated through pilot installations at partner institutions.

- **WP 4 – Usability, Documentation, and Community Adoption:** WP4 improves usability and supports adoption of the platform in the research community. The work package provides documentation, training resources, and structured user feedback mechanisms to ensure effective use of the platform in research and teaching.

Used methods and tools

- **Agile and feedback-driven development:** Software development follows an agile process supported by issue tracking, code review, and continuous integration. Structured user feedback and a quality management approach ensure traceability of changes, high software quality, and continuous alignment with user needs.
- **Privacy by design:** Data protection is implemented as a core architectural principle throughout the development lifecycle. The platform integrates role-based access control, audit logging, and anonymisation and pseudonymisation workflows. Development is carried out in close coordination with institutional data-protection and IT-security experts to ensure compliance with regulatory and institutional requirements.
- **AI-assisted development and documentation:** Modern AI-assisted software development tools (e.g. GitHub Copilot and GPT-based code assistants) will support refactoring, testing, debugging, and documentation. These tools serve as developer support systems; all outputs are reviewed and validated by the project team. Their use increases development efficiency and supports the timely consolidation and documentation of the platform.

WP 1 – Consolidation and Documentation

Objective: To consolidate, clean, and document the existing NORA code base to create a robust, modular, and maintainable foundation for long-term development and external collaboration.

WP 1.1 - Code Consolidation and Harmonisation: The NORA code base has evolved over several years and currently includes components written in different languages (PHP, JavaScript/Node.js, Bash). This work package will consolidate and harmonise these components to reduce complexity and improve maintainability.

- Gradual replacement and harmonisation of legacy PHP and Bash components into a coherent Node.js-based backend and modern TypeScript/JavaScript environment
- Code cleaning and simplification: Removal of obsolete or redundant functions, streamlining of logic, and improved consistency across modules.
- Modularisation of the system into clearly separated components (data management, workflow engine, AI interface, visualisation, etc.) with defined APIs
- Systematic code review and refactoring of critical components to improve readability, maintainability, and performance
- Identification and optimisation of performance bottlenecks, including improved asynchronous task handling and load balancing for multi-user operation
- Database optimisation including schema review, indexing, and harmonisation of metadata models linking imaging data, users, and processing results

WP 1.2 - Monitoring, Testing, and Continuous Integration: Reliable operation requires systematic monitoring and automated testing procedures. This work package will establish a unified quality control and testing framework.

- Implementation of integrated system monitoring (disk usage, memory consumption, process status)
- Automated maintenance mechanisms including cleanup of temporary files and database caches
- Development of a unified test suite including unit tests, integration tests, and regression tests
- Integration of continuous integration pipelines (e.g. GitHub Actions or GitLab CI) for automated builds and testing
- Implementation of systematic debugging procedures and stress testing using synthetic and real-world imaging datasets

WP 1.3 – Data Protection and Security Framework: NORA builds upon the existing data protection concept of the University Medical Center Freiburg, developed together with the institutional data protection officer and approved by the local ethics committee. Within this work package, the framework will be more tightly integrated into the platform architecture. These mechanisms ensure that privacy and security requirements are embedded directly into the platform architecture following the principle of *privacy by design*.

- Integration of GDPR-compliant data handling mechanisms into core platform services
- Implementation and consolidation of role-based access control and user permission models
- Establishment of audit logging and traceability for all relevant processing operations
- Integration of anonymisation and pseudonymisation workflows for imaging datasets
- Definition of secure default configurations and security-relevant system settings

WP 1.4 - Developer Documentation and Collaborative Development: Comprehensive documentation is essential for sustainable research software development and for enabling external contributions.

- Creation of developer documentation automatically generated from annotated source code, including API documentation and architecture descriptions
- Documentation of development guidelines and coding standards
- Definition of contribution workflows and collaborative development procedures
- Establishment of version-controlled documentation platforms (e.g. ReadTheDocs or GitBook)

Methods and Development Approach: Development will follow agile principles with iterative sprints, prioritised via a ticket and issue tracking system (e.g. GitHub Issues or Jira). A lightweight quality management system will be applied, including code review policies, versioned releases, and changelog-based traceability. A dedicated bug tracker will be used for structured error reporting and correction cycles. To improve code quality and developer efficiency, modern AI-assisted tools (e.g. GitHub Copilot, GPT-based code review assistants) will be used for static analysis, documentation improvement, and automated bug detection.

All development and testing steps will be fully version-controlled, ensuring transparent provenance of code changes, and will adhere to the principle of “data protection and privacy by design”, embedding compliance, access control, and security mechanisms directly into the development process.

Risks and Mitigations: The consolidation of a heterogeneous legacy code base may introduce temporary instability or incompatibilities with existing workflows. This risk will be mitigated through incremental refactoring, continuous integration testing, and preservation of backward compatibility during migration steps. Because the platform processes sensitive medical data, strict data protection and security requirements must also be maintained throughout development. Close coordination with the institutional data protection officer and adherence to the already approved data protection framework will ensure compliance. Automated testing, staged deployments, and systematic monitoring will further reduce the risk of regressions during architectural changes.

Expected Outcome: A consolidated, modular, and well-documented software architecture forming a stable foundation for further platform development, external collaboration, and long-term sustainability of the NORA infrastructure.

WP 2 – Interoperability and Integration

Objective: To establish interoperability between NORA and major imaging standards, toolboxes, AI frameworks, and other national research infrastructures (e.g. NFDI), enabling seamless data exchange, reproducible workflows, and integration with external hospital and research systems.

WP 2.1 - Integration and Harmonisation of Imaging and Clinical Standards: To ensure compatibility with diverse clinical and research systems, NORA will be extended and harmonised with the following key standards:

- Integration of DICOM DIMSE / DICOMweb / WADO for direct communication with PACS and imaging archives, enabling bidirectional image retrieval and metadata handling
- Integration of FHIR (Fast Healthcare Interoperability Resources) for structured exchange of patient and study metadata with hospital information systems
- Support for HL7 interfaces for connectivity with clinical workflow systems and transfer of patient and study identifiers
- Implementation of BIDS (Brain Imaging Data Structure) and NIfTI/JSON conventions to support reproducible research data organisation
- Development of connectors to external SQL databases enabling integration with institutional research data warehouses or project databases through configurable schema mapping and secure authentication
- Implementation of automated validation routines to verify standards compliance and prevent interoperability regressions

WP 2.2 - Container Interface for External Algorithms: To enable flexible integration of external algorithms and processing pipelines, a standardised interface for containerised

modules will be defined. This framework will make it straightforward to add new tools, including user-developed algorithms, and will ensure reproducibility across installations.

- Specification of standardised input/output folder structures and metadata schemas
- Definition of a common entry-point protocol for launching, monitoring, and logging container jobs
- Support for resource allocation, parameter passing, and result provenance
- Automatic registration of containerised modules within NORA with version and citation metadata

WP 2.3 - Integration of Imaging Toolboxes and AI Frameworks: Building on existing connectors, this work package will establish stable and tested adapters to widely used research toolboxes and AI frameworks.

- Integration of research toolboxes including SPM, CAT12, ANTs, FSL, FreeSurfer, Lead-DBS, and diffusion MRI tools such as MRtrix and DIPY
- Harmonisation of existing AI interfaces (e.g. U-Net and PatchWork) and extension to frameworks such as nnU-Net and TotalSegmentator
- Implementation of unified workflows for model training, inference, and evaluation
- Integration of a Hugging Face connector enabling access to curated AI models while ensuring local execution for data protection and reproducibility

WP 2.4 – Clinical text and LLM integration: To complement image-based workflows, NORA will support the integration of structured and unstructured clinical text such as radiology reports and annotations. All components will follow the same principles of modularity, traceability, and privacy-by-design as image-based processing modules.

- Development of interfaces for importing and structuring clinical text data
- Integration of modular LLM-based components for structuring and linking textual information with imaging data
- Implementation of traceable workflows connecting text-derived information with image-based analyses

WP 2.5 - FAIR-Compliant Metadata and Infrastructure Integration: To align with national and international research data infrastructures, NORA will implement mechanisms supporting FAIR principles and interoperability with initiatives such as NFDI.

- Implementation of FAIR-compliant metadata structures for datasets, algorithms, and workflows
- Support for persistent identifiers and version tracking of models and workflows
- Integration with external repositories and infrastructures through open APIs
- Development of import/export mechanisms supporting interoperability with external infrastructures and repositories

WP 2.6 - Evaluation and Feedback: Pilot sites and partner hospitals will conduct integration testing under real-world conditions.

- Deployment of interoperability interfaces at pilot installations

- Evaluation of compatibility, performance, and usability in clinical and research environments
- Iterative refinement of interfaces based on feedback from partner institutions

Methods and Development Approach: Development follows the agile methodology established in WP1 using iterative development cycles coordinated through issue tracking and peer review. Continuous integration pipelines will automatically build and test all connectors against reference datasets. Conformance testing tools (e.g. DICOM validators and BIDS validator) will be used to ensure standards compliance.

To improve development efficiency and code quality, modern AI-assisted tools (e.g. Codex, Claude etc) will support static code analysis, documentation generation, and integration stub creation. All code changes will be version-controlled to ensure full traceability.

FAIR principles (Findable, Accessible, Interoperable, Reusable) will guide the design of metadata structures and interfaces.

Risks and Mitigations: Interoperability work depends on evolving external standards and software ecosystems. Changes in external APIs or toolbox interfaces could introduce incompatibilities. This risk will be mitigated by implementing modular adapters and container-based integration layers that isolate external dependencies. Integration with clinical systems also requires careful handling of data protection and security requirements; therefore, all connectors will follow the privacy-by-design framework established in WP1. Early testing with pilot sites will allow potential interoperability issues to be identified and resolved during development.

Expected Outcome: A platform capable of interoperating with major clinical standards, imaging toolboxes, AI frameworks, and national research infrastructures, enabling reproducible and scalable imaging workflows across institutions.

WP 3 – Reproducible Deployment and Infrastructure Integration

Objective: To make NORA broadly usable through containerised local installations, cloud-ready configurations, and a secure online playground.

WP 3.1 – Containerised Distribution and Deployment Framework: NORA will be prepared for easy installation and maintenance at research institutions through modern packaging and deployment tools.

- Development of containerised distributions (Docker images) with versioned releases
- Creation of deployment profiles and configuration templates for typical research environments
- Automated installation scripts and configuration validation tools
- Implementation of secure default configurations and update mechanisms
- Validation of installation procedures across multiple operating systems and institutional infrastructures

WP 3.2 – Cluster and High-Performance Computing Integration: Many imaging research workflows require large-scale computational resources. NORA will therefore be extended to support efficient execution of workflows on high-performance computing (HPC) systems.

- Integration with common job scheduling systems (e.g. SLURM)
- Support for distributed workflow execution and parallel processing
- Resource allocation and scheduling mechanisms for compute-intensive tasks
- Optimisation of data transfer and storage strategies for large imaging datasets

WP 3.3 – Cloud Deployment Profiles: In addition to institutional clusters, many research environments increasingly rely on cloud computing resources. NORA will therefore provide deployment profiles supporting scalable cloud-based execution.

- Development of container orchestration configurations (e.g. Kubernetes)
- Support for scalable execution of processing pipelines in cloud environments
- Integration of monitoring and resource management tools
- Evaluation of deployment on selected cloud platforms and institutional cloud infrastructures

WP 3.4 – Pilot Research Installations: NORA will be deployed and validated at multiple partner hospitals and research institutions.

- Installation at partner sites with guided setup
- Training for local administrators and provision of standard operating procedures
- Collection of feedback on usability and performance for continuous improvement
- Preparation of install documentation and guidelines

WP 3.5 – Online Demonstrator Environment: A lightweight public version will provide a safe entry point for new users and serve teaching purposes.

- Demonstration of annotation, processing, and AI evaluation workflows
- Implementation of automatic cleanup, session reset, and anonymised usage logs
- Use as a training and outreach tool to attract new users

WP 3.6 – Governance and Sustainability: To ensure long-term operation, governance and maintenance structures will be formalised.

- Definition of an open-source governance model and release cycle
- Establishment of a core development and maintenance team at the University Medical Center Freiburg
- Exploration of funding and service models for continued support
- Publication of a sustainability and maintenance plan

Methods and Development Approach: An agile, feedback-driven process will guide development and deployment. Continuous integration pipelines and Docker-based packaging will ensure reproducible builds and deployments. Bug and feature tickets, together with structured feedback from pilot sites, will prioritise improvements in usability, scalability, and documentation.

Risks and Mitigations: Deployment across heterogeneous infrastructures (local installations, HPC clusters, and cloud environments) may introduce compatibility issues and performance variability. This risk will be mitigated through containerised distributions, standardised deployment profiles, and early validation at partner institutions. Integration with external computing infrastructures will be implemented incrementally and tested with representative workloads to ensure stability. Governance and sustainability challenges will be addressed by establishing a clear open-source maintenance model and long-term development responsibilities at the University Medical Center Freiburg.

Expected Outcome: A reproducible deployment framework enabling NORA to be installed and operated across institutional, HPC, and cloud environments, together with pilot installations, training resources, and governance structures that support sustainable adoption beyond the original development site.

WP 4 – Usability, Documentation, and Community Adoption

Objective: To empower non-expert users, promote best practices in medical imaging research, and build a sustainable user and developer community around NORA. While previous work packages focus on software architecture, interoperability, and deployment, this work package ensures that the platform can be effectively used and adopted by the research community. Particular emphasis is placed on user-centred design, documentation, and structured feedback from pilot installations and early adopters.

WP 4.1 – User Interface and Workflow Optimisation: The usability of NORA will be systematically improved through refinement of the user interface and common research workflows. These improvements will reduce the entry barrier for non-technical users while maintaining the flexibility required by advanced users and developers.

- Simplification of typical analysis workflows (data selection, annotation, processing pipelines, result exploration)
- Improvement of visualisation and quality-control tools for imaging and derived results
- Streamlining of navigation and dataset management interfaces
- Integration of usability feedback from pilot installations and early adopters

WP 4.2 – User Documentation and Tutorials: Comprehensive learning resources will ensure that new users can quickly adopt NORA for research and teaching. Documentation will include:

- Development of step-by-step tutorials for common research workflows
- Creation of illustrated user manuals and quickstart guides
- Provision of example datasets and reproducible workflow templates
- Production of short video introductions and interactive tutorials

WP 4.3 – Usability Evaluation and Feedback Integration: Structured usability testing will be conducted together with pilot research installations and early adopters. Results of these evaluations will guide iterative improvements in user interface design, workflow structure, and documentation.

WP2.2 – Container Interface for External Algorithms														
WP2.3 – Integration of Imaging Toolboxes and AI Frameworks														
WP2.4 – Clinical text and LLM integration														
WP2.5 – Metadata Library and FAIR Integration														
WP2.6 – Evaluation with Pilot Research Installations														
WP3.1 – Containerised Distribution and Deployment Framework														
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WP3.3 – Cloud Deployment Profiles														
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WP3.5 – Online Demonstrator Environment														
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WP4.1 – User Interface and Workflow Optimisation														
WP4.2 – User Documentation and Tutorials														
WP4.3 – Usability Evaluation and Feedback Integration														
WP4.4 – Training and Community Development														

Im diagram die Stellen abbilden, inkl der Eigenbeteiligungen, ggf milestones

Personnel required for the work programme: The implementation of the work programme requires a core team combining software engineering, biomedical imaging expertise, and clinical input. The personnel listed below represent the direct effort allocated to the work packages. Additional contributions from associated collaborators and partner institutions (see Section 5.2 to 5.4) - including consultation, testing, and feedback - will support the project but are not covered by the requested funding.

Position	Effort	Requested DFG Funding	Role
PostDoc1	50%	0%	
MD	50%	25%	
PhD1	100%	100%	
PhD2	100%	100%	

Requested funding: Requested funding: The University Medical Center Freiburg will contribute PostDoc1 and 25% clinician effort (MD) as institutional contributions to the project (see Section 5.9). Funding from the DFG LIS programme is requested for two full-time doctoral researchers (PhD1 and PhD2), or equivalent positions, and the remaining 25% of the MD position.

2.4 Relevance of sex, gender and/or diversity in the research project

The project itself does not specifically target sex- or gender-related questions, but NORA will enable such analyses by providing structured, standardised, and reproducible access to

large imaging datasets. The platform supports sex- and diversity-specific research through consistent metadata handling and cohort stratification. Recruitment and collaboration will follow equal-opportunity principles and promote diversity within the project team.

3 Project- and subject-related list of publications

*Works cited from chapters 1 and 2, both by the applicant and by third parties. Please include DOI/URL if available. A **maximum of ten** of your own works cited may be **highlighted**; font at least Arial 9 pt.*

[Text]

Irgendwo schreiben

There is no standalone technical publication dedicated to NORA; however, its scientific impact is evidenced by explicit acknowledgment in more than 180 peer-reviewed publications, spanning multiple imaging modalities, cohorts, and research questions. The following list represents an excerpt of these publications.

4 Supplementary information on the project context

Section 4 et seq. must not exceed 10 pages.

4.1 General ethical aspects

No specific ethical approval is required, as the project focuses on software development and research infrastructure. Ethical principles and data protection are core design elements of the NORA platform.

NORA supports the responsible use of medical imaging data and AI-based methods by ensuring transparency, traceability, and reproducibility of analyses. Data protection is implemented according to a privacy-by-design and privacy-by-default approach, with role-based access control, audit logging, and integrated anonymisation workflows embedded at the architectural level.

The platform already operates under an established data protection concept at the University Medical Center Freiburg, developed in cooperation with the institutional data protection officer and approved by the local ethics committee (EK-Freiburg 21-1215). This framework ensures GDPR compliance and will serve as the basis for further development and ethical governance throughout the project.

Ggf data privacy statement von officer anfügen

4.2 Considerations on aspects of ecological sustainability in the planning and implementation of the project

NORA actively supports ecological sustainability by enabling the efficient use of computing and storage resources in medical imaging research. Through smart cluster load management, resource allocation, and automatic cleanup routines, the platform helps reduce energy consumption and data waste. Development will follow energy-efficient computing principles and reuse existing infrastructure wherever possible.

4.3 Measures to meet funding requirements and handle project results

Details relating to the requirements stated in the relevant programme guidelines or call text under section I.2.2 ("Proposal requirements and funding conditions"), particularly with regard to handling results and data achieved within the project (measures for storage, long-term provision and dissemination)

Describe in detail measures which will ensure that the information infrastructures developed with DFG funding will continue to be reliable and, especially when it comes to consolidated technologies, sustainable after DFG funding expires.

All project results will be comprehensively documented, versioned, and made openly available in accordance with DFG guidelines. The NORA platform, including source code, documentation, test suites, and example workflows, will be released under an open-source license (e.g. MIT or GPL) via a public Git repository hosted and maintained by the University Hospital Freiburg.

To ensure sustainability beyond the funding period, NORA will be embedded into the existing institutional research infrastructure at UKF, where it is already actively used and maintained. A core development and maintenance team will remain in place, financed through institutional funds. The project will also establish a community-driven governance model, allowing external contributors to participate and ensuring continued evolution of the platform after DFG support ends.

All data, software components, and documentation will adhere to the FAIR principles (Findable, Accessible, Interoperable, Reusable). Metadata and workflows will be made available through recognised repositories (e.g. Zenodo, EMBL, OpenNeuro). Access rights, version control, and reproducibility will be managed transparently, ensuring the long-term reliability, openness, and sustainability of the developed infrastructure

4.4 Formal assurances

Formal assurance must be given that specific conditions of funding will be met. These depend on the funding programme and can include long-term provision of project results, compliance with specifically named standards, or the amount of the applicants' own financial contributions (see "Proposal structure" in the relevant programme guidelines or call text).

Copy the relevant assurances and declarations explicitly required in the programme guidelines.

Please confirm by stating the following phrase in the proposal that "publications resulting from the project and any relevant documentation will be available via open access, making them widely accessible for use by third parties" and that "the source code for the software developed under the project will be documented in accordance with the principles of open source and made available for use by third parties".

The applicants hereby confirm that publications resulting from the project and any relevant documentation will be available via open access, making them widely accessible for use by third parties and that the source code for the software developed under the project will be documented in accordance with the principles of open source and made available for use by third parties.

Long-term accessibility and sustainability of all project results will be ensured in accordance with DFG requirements.

5 People/collaborations/funding

5.1 Employment status information

For each applicant, state the last name, first name, and employment status (including duration of contract and funding body, if on a fixed-term contract).

Noch unklar wer Antragsteller: Marco, Elias, beide?

Problem Elias: Kann dann nicht aus den Mittel bezahlt werden.

Dr. Marco Reisert	since xxx
Dr. Elias Kellner	since xxx limited contract ...

5.2 Composition of the project group

List only those individuals who will work on the project but will not be paid out of the project funds. State each person's name, academic title, employment status, and type of funding. Please list separately the individuals paid by their institution(s) and those paid using other third-party funding (including fellowships).

The project is supported by an experienced interdisciplinary team of clinicians, imaging scientists, and software developers at the University Hospital Freiburg (UKF). The group has been working together for several years on the conception and implementation of the NORA platform and will contribute significant in-kind effort to this project. Details of personnel and infrastructural contributions are provided in Section 5.9. The following key contributors will participate in the project but will not be paid from DFG funds

Dr. Marco Reisert, Senior Scientist, UKF – institutional funding
 Dr. Elias Kellner, Senior Developer, UKF – xxxx funding
 Dr. Robin Schirrmeyer, Computer Scientist, UKF – xxxx funding
 PD Dr. Maximilian F. Russe, Clinician Scientist, UKF – institutional funding
 PD Dr. Alexander Rau, Clinician Scientist, UKF – institutional funding
 <externe>

This established group ensures continuity, interdisciplinary depth, and direct access to clinical and computational expertise essential for the project's success.

5.3 Institutions or researchers in Germany with which/whom you have agreed to cooperate on this project

If you will be pursuing your project jointly with researchers working in Germany and have shared responsibility for the conduct of the project, list the names of these individuals under co-investigators. The term co-investigator refers to individuals who are eligible to submit proposals but who neither request nor receive project funding. In addition, list the names of researchers in Germany with whom you will be collaborating on the proposed project but who will not share responsibility for the conduct of the project. Any such agreements must be attached to the proposal.

GUIDE-IT?

Tübingen - Prof. Dr.-Ing. Thomas Küstner

Essen - Forsting / Nensa (ggf sieht er sich als mit Konkurrenz), letter of support?

NUM - Racoon? Läuft über Mint Lesion Software

PC3

AIDA

BioPro

Konnektoren zu NUM/Racoon / OMI / NAKO ...

OMI Platform - Roman Stump / Elmar K

Charité - wer?

Ulm - wer?

5.4 Institutions or researchers abroad with which/whom you have agreed to cooperate on this project

If you intend to pursue the project in close collaboration with researchers or institutions based outside Germany, state the names of these individuals or institutions and please state whether

- the project will be conducted as part of an agreement between the DFG and a partner organisation, www.dfg.de/internationales/partner Please note the individual country information provided.

- Whether, and if so, to what extent your cooperation partners in a partner organisation are applying for DFG funding. A corresponding signed letter of commitment must be submitted in the attachment to your proposal.

Stanford - Sergios?

Basel ?

Harvard / Jogesh (Matze Jung)?

NYU

USA partners could explore commercial possibilities (US maybe more market than germany)

Maxim, Valerij, Jakob W fragen

5.5 Institutions and researchers with which/whom you have collaborated scientifically within the past three years

This information will help avoid potential conflicts of interest.

[Text]

5.6 Project-relevant cooperation with commercial enterprises

If applicable, please note the EU guidelines on state aid or contact your research institution in this regard.

None

5.7 Project-relevant participation in commercial enterprises

Please indicate if you are the owner of a commercial enterprise or a stakeholder in one (e.g. a director). If so, please state how your research project is linked to the company's production branch or activities. Please refer to DFG form 4.03, "Information on Work Results and Usage Rights in the Event of an Association with a Company".

Falls Elias

Dr. Elias Kellner is a shareholder of VEObrain GmbH, Freiburg, a company that develops and commercialises individual medical imaging and AI algorithms for clinical application. VEObrain does not develop or operate research or information infrastructures.

The proposed project NORA is an open-source, non-commercial research infrastructure focused on data management, interoperability, and reproducible workflows. There is no competition or overlap between VEObrain's commercial activities and the objectives of this project; the activities are complementary in scope.

All results of the DFG-funded project remain under the control of the University Medical Center Freiburg. No project results or usage rights will be transferred exclusively or preferentially to VEObrain GmbH. Upon grant acceptance, a formal usage agreement between VEObrain and UKF will be concluded to ensure full compliance with DFG regulations.

5.8 Other submissions

List any funding proposals for this project and/or major instrumentation previously submitted to a third party.

None

5.9 Financial contributions

State how much funding and what equipment and staff will be contributed to the project by the applicant(s). This information must be provided in detail in the proposal. Please note the instructions in the programme guidelines and use the structure and order given in the section "Requested modules/funds" to list the information. Please indicate explicitly if the equipment will be available on site for the project. If you are applying for instruments that are available at your institution, but are not at the project's disposal, please explain why this is the case in the section "Requested modules/funding".

Applicants are expected to make a reasonable financial contribution to the project, e.g. in the form of staff and direct project costs. For projects aimed at the implementation or consolidation of information infrastructures, the contribution is expected to be significantly higher than for experimental projects.

The project builds on a long-standing institutional commitment at the University Medical Center Freiburg. The core development team has already invested significant time and effort in creating and maintaining the NORA platform without external funding and will continue to do so throughout the project. This ongoing in-kind contribution ensures continuity, technical stability, and deep integration with clinical workflows.

The institution will finance a **50% postdoctoral position and two 25% clinicians** for the entire project duration. In addition, the team will have full access to UKF's extensive computing and data infrastructure, including 12 servers, 14 GPUs, and more than 20 TB RAM, as well as existing databases, PACS connections, and research networks. These resources represent substantial direct institutional costs already covered by the university and are fully available for project use.

To enable secure online access and external collaboration, a limited portion of the requested funding will cover dedicated demonstration servers and specialised data-security consulting (e.g. server hardening, GDPR compliance). No new instrumentation is requested.

Overall, the Medical Center Freiburg provides a comprehensive and well-equipped environment for development and testing, exceeding the expected level of own contribution for an infrastructure-oriented project and underlining the institution's strong long-term commitment to NORA.

Ggf noch Datenschutzbeauftragter Spieler eigenanteil

6 Requested modules/funds

Explain each item for each applicant (stating last name, first name). Structure according to the corresponding programme guidelines or call text.

6.1 Requested modules/funds

Under this funding programme it is possible to apply for the basic module, potentially in conjunction with the "Project-Specific Workshops" module. For more details, please see the respective guidelines for each module.

1. Basic Module Use the basic module to request funding for direct project costs, project-specific staff and instrumentation necessary to carry out the project. www.dfg.de/formulare/52_01

2. Module Project-Specific Workshops

If you would like to conduct workshops as part of your project, you may request funding to help you do so. Please note that the "Project-Specific Workshops" module cannot be submitted separately but only in conjunction with the proposed project.

[Text]

Basic Modul: Sachmittelbeihilfe

Workshop-Modul würde Sinn machen, aber dann orga usw -> wer macht das?

7 Annex / Other information

Parts of this proposal were prepared with editorial assistance from ChatGPT (OpenAI). The ideas, structure, and all scientific content were developed, reviewed, and verified by the authors.

Part C: Attachments

Additional information

The following attachments can be included with a proposal; however, the proposal must also be fully comprehensible without reading the attachments:

- Needs analysis
- Environment analysis
- Concept for permanent safeguarding incl. institutional letter of assurance
- Letters of support and letters of intent

If a proposal is aimed at the further development or advancement of established research software, the following additional attachments can be included; however, the proposal must also be fully comprehensible without reading the attachments:

- Justification of relevance
- Proof of usage rights for third-party code, if applicable

If the proposal is submitted by members of an academic information infrastructure facility, it must be accompanied by a declaration on the part of the management (see DFG form 12.141) stating the following (where applicable, depending on the phase):

- the financial contribution required within the scope of the programme will be made;
- the results of the project will be supported once DFG funding has expired.

Vordruck genügt, kein extra schreiben aufsetzen

www.dfg.de/formulare/12_141

Need Analysis

<GUIDE-IT ergebnisse?>

A systematic needs assessment was carried out through literature and internet searches, a review of existing imaging research platforms, and user feedback from research groups at the University Hospital Freiburg and partner institutions. The analysis revealed that, although numerous tools exist for specific aspects of medical image analysis—such as XNAT, MITK, 3D Slicer, CBICA CaPTk, JIP, Flywheel, Aicura, QMENTA, and Orthanc—these systems typically address only isolated components of the research workflow. Most are either commercial and closed-source, or lack integration of data management, annotation, AI processing, and reproducible evaluation within a single, open, and locally deployable framework.

The comparative analysis (see table below) confirms that no open and privacy-compliant infrastructure currently combines data handling, algorithm integration, scalable processing, and collaborative image interaction in one unified system. In particular, interactive data viewing, annotation, and collaborative quality control are either missing or implemented only through external plug-ins or desktop tools. In contrast, NORA provides a fully integrated, web-based environment that allows researchers to view, annotate, and process imaging data interactively and collaboratively, directly within the platform. This deep integration and accessibility make NORA unique among open and non-commercial systems.

Platform / Provider	Open Source	Local Installation	Data Mgmt / PACS	Data viewing / annotation	AI Integration	Scalability / Cluster Use	Focus / Limitations
Orthanc	✓	✓	✓	✗	✗	✗	Lightweight open-source PACS server; ideal for image storage and DICOM interoperability but lacks research

Platform / Provider	Open Source	Local Installation	Data Mgmt / PACS	Data viewing / annotation	AI Integration	Scalability / Cluster Use	Focus / Limitations
							workflow, annotation, or AI integration features
XNAT (Wash. Univ.)	✓	✓	Limited	✗	✗	Limited	Strong for data storage/metadata; limited AI/annotation
MITK (DKFZ)	✓	✓	✗	✓	✓	✗	Viewer/interaction toolkit; not a full research environment
3D Slicer (BWH/Harvard)	✓	✓	✗	✓	Partial (plugins)	✗	Desktop-centric; lacks central data management/web
CBICA CaPTk (UPenn)	✓	✓	✗	Partial (plugins)	✓	✗	AI pipeline demos; not web/scalable; focus on cancer
Flywheel.io	✗	✗	✓	Partial	✓	✓	Commercial, subscription; cloud-first (privacy/sovereignty issues)
Aicura Medical	✗	✗	✓	Partial	✓	✓	Commercial; clinical deployment focus, closed
QMENTA	✗	✗	✓	Partial	✓	✓	Closed cloud platform
MeVisLab (Fraunhofer MEVIS)	✗	✓	✗	✓	Partial		Modular development framework; no web access, data management, or collaborative functionality
JIP (DKTK/DKFZ, Heidelberg)	—	✓	✓	Partial (plugins)	✓	✓	Federated, multi-site platform across DKTK; access primarily for participating clinics; public code availability not indicated
MONAI (NVIDIA)	✓	Partial (requires integration)	✗	Partial (plugins)	✓	✓	Excellent AI training and deployment framework; lacks data management, web interface, and workflow integration; complements rather than replaces NORA
NORA (proposed)	✓	✓	✓	✓ (fully integrated web-based, collaborative environment)	✓	✓	Open, locally deployable end-to-end research infrastructure (management, annotation, AI, reproducibility)

Table: Overview of Existing Imaging Research Platforms Compared to NORA

<Fehlen noch welche>?

<weitere zufügen: Mevis, ...?>

In Germany alone, there are approximately 40 university hospitals and more than 200 academic medical research institutions actively conducting imaging-based research. Within each hospital, typically 10–20 departments (radiology, neurology, oncology, cardiology, etc.) perform independent image analyses, representing a potential user base of 500–1,000 research groups nationwide. On a European scale, the need is substantially greater: more than 1,000 institutions are engaged in imaging research, and globally, over 100,000 publications per year now include keywords such as “medical imaging,” “AI,” or “radiomics.”

Particularly in the context of large-scale cohort studies such as the German National Cohort (NAKO), the UK Biobank (UKBB), and similar population-based imaging projects, the need for scalable, standardised, and reproducible analysis tools is becoming critical. These studies generate petabyte-scale imaging datasets that require consistent annotation, quality control, and AI-assisted analysis — tasks that demand precisely the type of integrated, high-throughput infrastructure NORA provides.

Feedback from pilot users and partner hospitals confirms the strong and growing demand for an open, standards-based, and locally deployable imaging research infrastructure that ensures data protection compliance (GDPR) while supporting collaboration and AI-driven workflows. NORA directly meets this need, bridging the gap between hospital environments and large-scale research and forming a foundation for sustainable, FAIR, and interoperable medical imaging research at national and international levels.

Concept for Permanent Safeguarding

After the funding period, NORA will be permanently maintained within the institutional IT and research infrastructure of the University Hospital Freiburg. A core maintenance and development team, supported by institutional funding, will ensure ongoing stability, updates, and compliance with security and data-protection requirements. The software and documentation will remain open source and archived via institutional repositories (GitHub and Zenodo). A community-based governance model will coordinate external contributions and ensure transparent development. In addition, the hospital's commitment to long-term hosting and maintenance will be formalised through an institutional letter of assurance, ensuring the sustainability and accessibility of NORA well beyond the project duration.

Justification of Relevance

The accompanying Needs Analysis clearly demonstrates a substantial gap between the increasing demands of imaging-based research and the availability of suitable open infrastructures. Existing tools and commercial platforms cover only isolated aspects of the workflow or impose restrictions that limit accessibility, scalability, and reproducibility.

NORA directly addresses this gap. It is already a proven, widely used platform at the University Hospital Freiburg, combining data management, annotation, processing, algorithm integration, and evaluation within one coherent system. The proposed further development will transform NORA into a sustainable, interoperable infrastructure accessible to the broader research community.

Given the lack of market-driven solutions in this area, the development of such infrastructures must be supported through public funding. NORA exemplifies a community-driven, open-source approach that aligns with the DFG's goals for FAIR, transparent, and reusable digital research environments. It will serve as a model for reproducible and trustworthy AI in medicine, with long-term benefits for both national and international research communities.

Proof of usage rights for third-party code

All third-party components integrated into NORA are licensed under widely used, permissive open-source licenses such as MIT, Apache 2.0, or BSD, which allow unrestricted academic use, modification, and redistribution. Medical imaging libraries such as DCMTK and OpenJPEG are similarly distributed under BSD-style or Apache-compatible licenses. For external research toolboxes accessed through NORA (e.g., SPM, FSL, ANTs, FreeSurfer), their respective license terms apply; NORA only interfaces with these tools and does not redistribute their code. No third-party components with restrictive or incompatible licensing terms are included.

Third-Party Components and License Types Used in NORA

Component / Library	Type / Use	License	Notes
Node.js ecosystem packages (glob, extend, chokidar, strip-json-comments, uid-number, nodemon, notify.js, q.js, xregexp)	Server-side utilities, file handling, runtime support	MIT / Apache 2.0 / BSD	Fully permissive, free for reuse and redistribution
daikon.js	DICOM parsing	MIT	Open and redistributable, widely used in imaging
jQuery	Web UI utilities	MIT	Long-standing permissive open-source license
Chart.js	Visualisation charts	MIT	Free for academic and commercial use
jsPDF	PDF generation	MIT	No restrictions on integration or redistribution
Font Awesome	Icon set	MIT (Free version)	Redistributable; attribution preserved
Alertify.js, Dragster.js	UI interaction components	MIT	Fully open-source
gzip, zip libraries	Compression	BSD / MIT	Permissive, widely used
OpenJPEG	JPEG2000 decoding	BSD-like / Apache 2.0	Open-source and freely redistributable
DCMTK	DICOM networking and file handling	BSD-style	Permits free use, modification, and redistribution
MySQL Connector	Database access	GPL / LGPL depending on version	No license conflict; used without static linking

External Toolboxes (not redistributed, only interfaced)

Toolbox	License	Notes
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SPM / CAT12	Free for academic use, MATLAB required	Users must hold their own MATLAB license
FSL	Oxford Academic License	Free for non-commercial research
FreeSurfer	FreeSurfer License	Free for research; redistribution allowed
ANTs	Apache 2.0	Fully open and permissive
Lead-DBS	GPL	Compatible since NORA does not embed code
nnU-Net, nnU-Net-interactive	Apache 2.0 / MIT	Fully open-source
TotalSegmentator	Apache 2.0 / MIT	Fully open-source
PatchWork	Apache 2.0 / MIT	Fully open-source
Hugging Face models	Vary (MIT, Apache, etc.)	Users must respect the model-specific license; NORA only provides connectors